

Omkar Arvind Kottawar

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SUMMARY

Computer Science undergraduate with experience building data-driven software using Python, Java, and SQL through open-source contributions and end-to-end projects. Experienced in backend development, data processing, and designing analytical pipelines that transform raw data into actionable insights. Strong foundation in software engineering with an interest in data analytics, scalable systems, and solving complex real-world problems.

EDUCATION

VIT Bhopal University

B.Tech. in Computer Science and Engineering - CGPA: 9.03/10

Sehore, MP

Sep 2023 – May 2027

Datta Meghe World Academy

Higher Secondary Certificate - Percentage: 81.4%

Airoli, MH

Aug 2023

Datta Meghe World Academy

Senior Secondary Certificate - Percentage: 95%

Airoli, MH

Aug 2021

EXPERIENCE

Software Engineering Intern

March 2026 – July 2026

FOSSEE, IIT Bombay

Mumbai, MH

- Refactored CAD generation modules within the OSDAG codebase, improving modularity, readability, and long-term maintainability.
- Debugged complex software workflows, resolved generation inconsistencies, and improved reliability across multiple structural design configurations.
- Collaborated using Git-based development workflows while enhancing documentation and reducing onboarding effort for future contributors.

PROJECTS

MiniDB — Relational Database Engine | Java, Swing, OOP

2026

- Designed and implemented an in-memory relational database supporting schema creation, CRUD operations, and constraint enforcement using object-oriented design principles.
- Implemented primary-key uniqueness, NOT NULL validation, and type checking while maintaining schema integrity through a modular architecture.
- Built an interactive Java Swing desktop application for database creation, table management, record insertion, updates, deletion, and real-time data visualization.
- Structured the system using reusable components and separation of concerns, enabling future extensibility for SQL parsing and persistent storage.

Context Handoff Protocol — AI-Powered Team Handoff Briefs | Python, FastAPI, HDBSCAN, LLMs

2026

- Built a full-stack pipeline that ingests cross-platform team signals (Slack, GitHub, Notion, email) and synthesizes structured, AI-generated handoff briefs for engineering and product teams.
- Engineered a modular four-stage pipeline (ingestion, semantic clustering via HDBSCAN, staleness scoring, LLM synthesis) with provider-agnostic embedding and LLM interfaces swappable via configuration alone.
- Designed a FastAPI backend with async SQLAlchemy/SQLite persistence and a responsive single-page frontend, deployed on HuggingFace Spaces with pluggable connector architecture for real and synthetic data sources.

ReviewSense — Customer Review Analytics Platform | Python, Pandas

2025

- Built a review analytics application that processes customer reviews and generates interpretable satisfaction scores through a modular processing pipeline.
- Implemented review ingestion, feature extraction, scoring, and CSV export with emphasis on reusable software components and maintainable code.
- Designed the application for extensibility, enabling future integration with dashboards and large-scale datasets.

TECHNICAL SKILLS

Programming: Java, Python, SQL

Backend: FastAPI, REST APIs, SQLAlchemy, Async Programming

Databases: SQLite, Relational Database Design

AI/ML: PyTorch, NumPy, Pandas, HDBSCAN, LLM Integration

Tools & Platforms: Git, Java Swing, Streamlit, Hugging Face Spaces

Cloud: Oracle Cloud Infrastructure

CERTIFICATIONS & TRAINING

Oracle Cloud Infrastructure AI Foundations Associate, Oracle University (2025–2027)

The Bits and Bytes of Computer Networking, Google (2025)

Machine Learning Specialization, DeepLearning.AI & Stanford Online (2026)

Retrieval-Augmented Generation (RAG), DeepLearning.AI (2026)